

Virtual Reality Antarctic Weather Station Repair for Informal STEM Learning

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Figure 1: Screenshots from the developed experience, “On the Ice: Weather Station”. The image on the left shows a puzzle that the player needs to solve in an outdoor space, while the images on the right show puzzles for the user to solve in an indoor space. These puzzles provide a gamified way to learn about the real-life science happening in Antarctica surrounding weather stations.

ABSTRACT

Developed in collaboration with polar science experts, this virtual reality game immerses users in the life of a scientist maintaining weather stations in Antarctica. Designed for the Meta Quest 3, the game challenges players to complete realistic repair tasks reimagined as engaging tactile puzzles. As players tackle these challenges, a virtual assistant provides guidance, not only assisting with game-play mechanics but also offering valuable insights into the real-world science behind their actions. This interactive experience aims to captivate players aged ten and above, offering a unique blend of education and entertainment. A preview of the experience can be found at <https://www.youtube.com/watch?v=cQpikAcYwek>.

Index Terms: Virtual Reality, Informal Learning, STEM Education, HMD

1 INTRODUCTION

Embodied learning has shown significant promise for teaching complex STEM topics [6], as well as increasing empathy and perspective-taking [4]. Virtual Reality has shown to be a powerful tool for utilizing these affordances for informal learning environments [2, 7, 8, 9]. This project aims to create a playful experience in which the player embodies a polar researcher whose job it is to maintain weather stations in Antarctica. The goal of this work is to create an informal learning environment in which the player will gain a better understanding of polar research, researchers, and the impact of polar weather on climate.

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2 METHODS

To achieve these goals, a playful virtual reality (VR) game was created in the Unity game engine. The experience utilizes two-handed interaction and room-scale tracking and was designed for those ages eight and older, with a particular focus for those unfamiliar with VR.

In the virtual experience, the player is transported to a remote locations in Antarctica, where they are tasked with repairing a collection of weather stations. The player is tasked with completing different real-life repair tasks, which are re-invented in the form of various tactile puzzles. Examples of these puzzles are shown in Figure 1. For each puzzle, the player must identify the components that are in disrepair, fix them in your mobile repair workshop, and then reinstall them. The player must work quickly to avoid getting stuck in a dangerous windstorm or left behind.

While working through the puzzles for each station, the player is provided guidance through a virtual aid, Argo, who exists as a personified AI handheld control box. The player interacts with Argo via a red button and while Argo interacts with the player verbally (see Figure 2). Throughout the experience, the player is required to bring Argo along with them on their journey, during which Argo provides the player with guidance on puzzle and game-play mechanics. Additionally, Argo provides the player with information regarding their characters backstory and factual information about the endeavors of Antarctic researcher and its larger significance. When not held in the player’s hand, Argo will snap to a dedicated surface nearby within arms reach (see Figure 1).

For the player to successfully complete the game, they need to help restore a network of weather stations that are linked via satellites before they trapped in a storm. To facilitate different types of learning experiences, the game was split up into a series of game-play loops, each meant to be experienced in approximately 10 minutes. These loops were designed to reinforce the targeted learning objectives.

3 LEARNING OBJECTIVES

In the game, players are meant to learn about the unique challenges of scientific research in Antarctica. They’ll encounter the stark

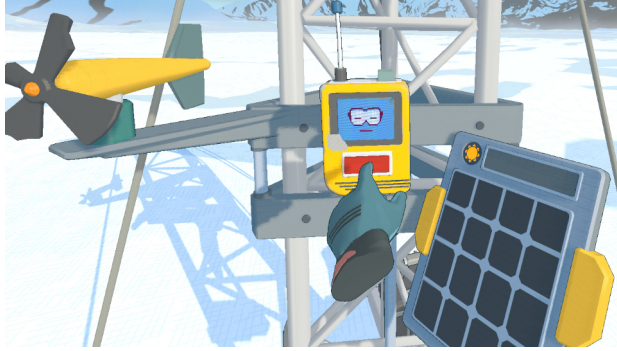


Figure 2: Shows a user interacting with Argo during a mission to repair a weather station.

beauty of the Antarctic landscape – vast, snowy, and rocky – while experiencing firsthand the extreme weather conditions. These harsh realities dictate when, where, and how research can be conducted.

The game introduces players to the diverse international community of scientists drawn to this icy continent, each with their own cultural backgrounds and languages. Players learn about the varied modes of transportation used to navigate Antarctica, from sturdy research vessels to specialized aircraft, with choices influenced by weather and distance.

A critical aspect of the experience is providing knowledge around the scientific process of weather observation. Players learn how scientists record weather patterns over time and across different locations to make predictions. Players discover the network of Automatic Weather Stations (AWS) scattered across the continent, equipped with advanced technologies to gather crucial data like wind speed, direction, and temperature.

In the experience, players learn how this data is transmitted to polar-orbiting satellites, contributing to a deeper understanding of the Antarctic climate and its changing patterns. The game also highlights the practical challenges of maintaining these vital stations, from on-site repairs to the need to bring equipment indoors for more complex servicing. Through this immersive experience, players gain a deeper appreciation for the dedication of scientists working in Antarctica and the crucial role of technology in understanding our planet's changing climate.

4 RESULTS

The developed experience, “On the Ice: Weather Station”, is currently available on the Meta App Lab for free and compatible with the Meta Quest 3 head mounted display (HMD). The experience is being deployed in public libraries as a part of custom curriculum and has been shown at STEM outreach events. While the experience has yet to be formally tested, the informal responses by outreach participants and play-testers has been very positive. Future work will aim to study the impact of this experience on informal learning.

Additionally, the experience is built using the Open Game Data framework [3]. This framework provides support for capturing all user actions which, after being anonymized, are shared publicly on a monthly basis. These datasets are currently being utilized for research which aims to better understand the user experience and predict player outcomes in means of developing real-time intervention strategies.

5 CONCLUSIONS

This work presents a virtual reality experience aimed at creating an informal learning space around Antarctic research. Through the

game, the player will better understand how polar research is performed, the impact of science gathered from this regions and the diversity of individuals who are participating in these research endeavors. Future work will aim to better understand the impact of the developed experience, with a particular focus on STEM identity [1], STEM interest [5] and domain knowledge.

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